

Trust in Robotics: A Two-Way Street

Charlotte Albert

*Dept. of Electrical and Computer Engineering
Queen's University
23qd12@queensu.ca*

Abstract—This paper investigates the idea that human perception of a robot can be improved through a reciprocal trust-building interaction. We present an architecture for a non-anthropomorphic robotic arm that expresses a gradual transition from fear and avoidance to comfort and affiliation contingent on user behavior. Inspired by dogs, the robot expresses emotional state through behaviors including gaze, shaking, head tilting, posture, orientation, bouncing, and dancing. By framing trust as a relationship that develops over time rather than an intrinsic principle of the robot, this work explores a new direction for strengthening human-robot connection.

Index Terms—Human-Robot Interaction, Reciprocal Trust, Social Robotics, Non-Anthropomorphic Robots

I. INTRODUCTION

How does trust define human-robot interaction (HRI)? This question is not novel. Public trust in robots has become increasingly important as they are integrated into daily life in domains such as education, healthcare, and transportation. For seamless integration into society, robots have to be reliable, likeable, and trustworthy, as these characteristics directly influence users' willingness to accept, rely on, and cooperate with robotic systems [1]. Researchers have changed aspects such as personalities, appearances, competence, gaze, and communication styles in an effort to gain users' trust. We hypothesize that a missing component is an explicit trust-building process from the robot's perspective. Rather than simply appearing trustworthy, the robot actively develops trust in the user through their interactions.

We ask: How does **the process** of building trust between a robot and user affect its likeability, perceived intelligence, and ability to connect? Relationships and bonds are dynamic; we theorize that a robot's immediate comfort may create a sense of superficiality, thereby causing unease. In contrast, a robot that gradually develops trust through interaction may foster a stronger perception of relational depth and meaningful engagement.

This work presents the development of a dog-inspired robotic character deployed on a non-anthropomorphic (NAR) robot. Gradual development of trust is displayed through its behavior, beginning with shaking, cowering, and cautious observation of the user, before transitioning to a relaxed posture and affiliative idle movements as trust is established. To gain the robot's trust, the user must move in a non-threatening manner. The selected NAR platform is the Universal Robotics UR5e manipulator for this interaction (see Figure 4 in VIII). Because the UR5e lacks a head or eyes, we use the Wrist 3 joint as a surrogate for gaze. By orienting Wrist 3 toward

the user, the robot maintains visual attention cues that are critical for coordinating social interaction and communication [2]. By requiring users to modify their behavior in response to the robot's initial wariness, trust is earned through reciprocal behavior rather than granted immediately.

Beyond improving human-robot rapport, reciprocal trust building may also lend itself to education in trust and boundaries. Social robots have been previously researched as therapeutic tools for kids with Autism Spectrum Disorder (ASD) and are found to be comforting due to their predictability and stylization [3]. Although this work does not evaluate therapeutic outcomes, the proposed architecture may provide a foundation for future systems to help children with ASD practice interpreting non-verbal cues and adapting their own behavior during social interactions.

II. RELATED WORK

An individual's perception of another strongly influences the quality of social interactions. In HRI study, Christoforakos et al. [4] found that trust in robots increases when people perceive them as competent and warm, particularly when they attach human-like qualities to the robot. Competence is attributed to agents when they can accurately perform tasks, while warmth is associated when the agent appears to have benevolent intentions. A key factor in associating these qualities is gaze. Early work from Kendon [2] points to gaze patterns in conversation and how critical it is for effective communication. In human-human interaction (HHI), direct gaze increases reliability and trustworthiness [5], with similar results found for human-animal interaction (HAI), where animal gaze, orientation, and posture coordinate interactions as these cues elicit human response [6]. In HRI, robotic gaze can increase perceived agency, even without realistic appearance [7].

Anthropomorphic appearances allow for mimicry of human facial features, reactions, and mannerisms. However, imperfect attempts at human-like appearance in robots can result in user unease, also known as the uncanny valley [8]. To avoid this effect, robotic behaviors can be explored with the use of NARs. El-Helou and Pan [9] create a socially engaging robot using a UR5e robot arm manipulator to explore the effects of gaze, posture, and movement on participant perception. The results show that attentiveness (looking at the user) increases feelings of warmth, competence, and intention, while high arousal (very dynamic movement) can increase feelings of disturbance.

Animal-inspired behavior provides an alternative for social-emotional engagement. Dogs express mood and behavior

through posture, orientation, gaze, movement, and proximity, which allows humans to infer their emotional state [10]. Since these behaviors do not rely on facial expression and are familiar to many, they can be used to translate expression in a NAR. In a study by Gásci et al. [11], the authors used a PeopleBot robot to exhibit dog-inspired emotional behavior and found that participants could successfully identify the robot's intended emotional state.

In addition, robots who can accurately perceive, interpret, and engage with participants have better success in connection [12]. Effective interaction requires the robot to anticipate and react to human action. When users see this behavior, they are more likely to attribute intentionality, competence, and social awareness to the robot [13].

III. OVERVIEW OF THE APPROACH

Guiding the development of the architecture was the definition of trust from Schilke et al. [14]: “The willingness of an entity (the trustor) to become vulnerable to another entity (the trustee). In taking this risk, the trustor presumes that the trustee will act in a way that is conducive to the trustor’s welfare despite the trustee’s actions being outside the trustor’s control.” To apply the above definition, we reference Maslow’s Hierarchy of Needs [15] as a framework for defining the robot’s progression towards trust. The first tier, *Physiological Needs*, is met for the robot, allowing it to focus on its connection with the user. Thus, the development of the relationship starts with the robot seeking the second tier, *Safety Needs*, toward the third tier, *Love and Belongingness*, where the robot demonstrates affiliation with the user.

The progression is detailed by the three Stages of Interspecies Relationship Development by Kieson [16]. Stage one centers on physical safety and the foundations of mutual communication. In this stage, trust is fragile; one must allow the trustor copious space and demonstrate consistent behavior. “Mutual language” is established through predictability and being nonthreatening. Stage two focuses on psychological safety and exchanges. During this stage, interactions become more exploratory and allow for social-emotional engagements. If behaviors are inconsistent or threatening, the trust built can be broken. Stage three centers on emotional bonding and seeking safety in partnership. The interactants can be physically and psychologically vulnerable. There is now an assertion of safety in the company of one another. In this work, these stages provide a structure for the robot’s trusting transition from avoidance to affiliative.

To implement the progression, we seek to develop a lifelike identity inspired by dog behavior. The selected behaviors correspond to different stages of the trust-building process described above. During initial interactions, dogs may tremble, shiver, pant, and monitor potential threats when experiencing stress or uncertainty [17]. These behaviors communicate fragility and wariness, aligning with Stage one, where physical safety and predictable communication must be established before trust can develop. Once a presence is perceived as non-threatening, dogs may approach and explore their environment.

Curious behaviors, such as head tilting or changing orientation, communicate interest and engagement, comparable with Stage two where social exchange develops. After investigation and a person is deemed trustworthy, positive affiliative behaviors, such as play and tail wagging, communicate comfort and social engagement [18]. These behaviors correspond with Stage three, where interactants develop emotional bonding and seek safety in partnership. To replicate this with a UR5e, we create a bouncing and dancing motion.

IV. SYSTEM DESIGN

The proposed architecture consists of four primary components: user perception, activity evaluation, trust estimation, and behavior generation. The user skeleton is tracked through an Intel RealSense D435 Depth Camera set adjacent to the robot. This information feeds into the activity engine, determining user activity through an *activity score*. The score determines the current trust level and appropriate behavior(s) in the robotic character’s trust engine. Behaviors are appropriately defined and layered within the robot movement publisher.

A. Activity Engine

The activity engine leverages the method introduced by Pan et al. [19], which is used to identify significant stimuli within the environment. Components such as distance, velocity, and raised hand increased the *curiosity score* of their robotic character, which influenced the target of its gaze.

The activity score, Φ , represents user activity perceived by the robot and is defined in Equation 1. The score is composed of six normalized components: head distance (D_{head}), hand distance (D_{hands}), velocity (V), acceleration (A), and binary indicators representing whether the right (I_{right}) or left (I_{left}) hand is raised. The user with the highest activity score becomes the robot’s current subject of interest.

$$\Phi = D_{\text{head}} + D_{\text{hands}} + V + A + I_{\text{right}} + I_{\text{left}} \quad (1)$$

Each component is normalized by dividing by the maximum measured value observed during operation. 0 represents the least threatening behavior: keeping a far distance, keeping hands low, moving slowly, and steadily. These were chosen for their correspondence with trustworthy behavior, such as moving slowly, predictably, and consistently, and keeping a low profile [18], [20].

B. Trust Levels

Based on the three stages of Interspecies Relationship Development described in Section III, we define a six-level trust progression to represent the robot’s transition from avoidance to affiliation. Trust levels 1a and 1b correspond to Stage one (physical safety), levels 2a and 2b correspond to Stage two (psychological safety and social exchange), and levels 3a and 3b correspond to Stage three (emotional bonding and partnership). Stage subdivision allows for more nuanced change. Figure 1 shows the number range, corresponding to the value of T , for each trust stage.

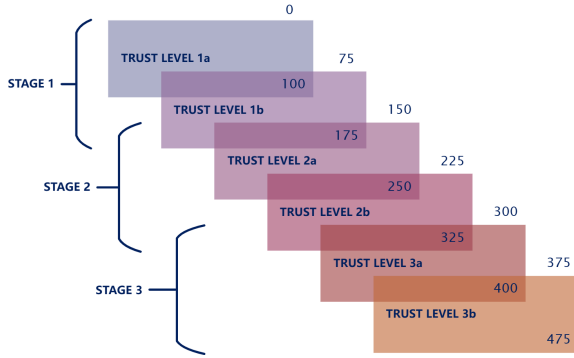


Fig. 1: Trust Levels.

Each trust level has a numerical range of 100, with slight overlap between levels dedicated for transitions. Since trust develops given the history of the interactions, the trust value, T is computed using Equation 2.

$$\begin{aligned}
 T(t) = & r_{\text{trust}}(t - t_0) \\
 & - \left(w_{\text{dist_head}} I(d_{\text{head}}, \tau_{\text{head}}) \right. \\
 & + w_{\text{dist_hand}} I(d_{\text{hand}}, \tau_{\text{hand}}) \\
 & + w_{\text{vel}} I(v, \tau_{\text{vel}}) \\
 & + w_{\text{accel}} I(a, \tau_{\text{accel}}) \\
 & \left. + w_{\text{hand_above}} (h_{\text{right}} + h_{\text{left}}) \right)
 \end{aligned} \quad (2)$$

The term $r_{\text{trust}}(t - t_0)$ is a global variable multiplied by the current duration to model the accumulation of trust over time. The global variable is adjustable depending on the desired duration of the interaction. Each trust level configures the robot's behavior by adjusting base position, activity thresholds, activity weights, and movement parameters.

Base Position: Base position describes where the robot orients itself. As trust level increases from Stage 1 to Stage 2, the base position is directed toward the user. As trust level increases in Stage 3, the posture becomes less timid and more comfortable. Orientation changes can be referenced in Figure 2 and a side view of posture changes can be seen in Figure 3.

Thresholds: From Equation 2, I represents an indicator variable. Each trust level has a different threshold, τ , for which the component will be subtracted from $T(t)$, the final trust value. Thresholds were manually tuned during development to produce a gradual transition and prevent abrupt behavioral change. As trust level increases, thresholds become less restrictive, representing reduced fragility.

Weights: The weights, ω , determine the magnitude of the penalty associated with each behavioral component. Components that are more likely to reduce trust, such as acceleration, distance, and raised hands, are assigned larger weights. As trust progresses, weights decrease to represent increased tolerance

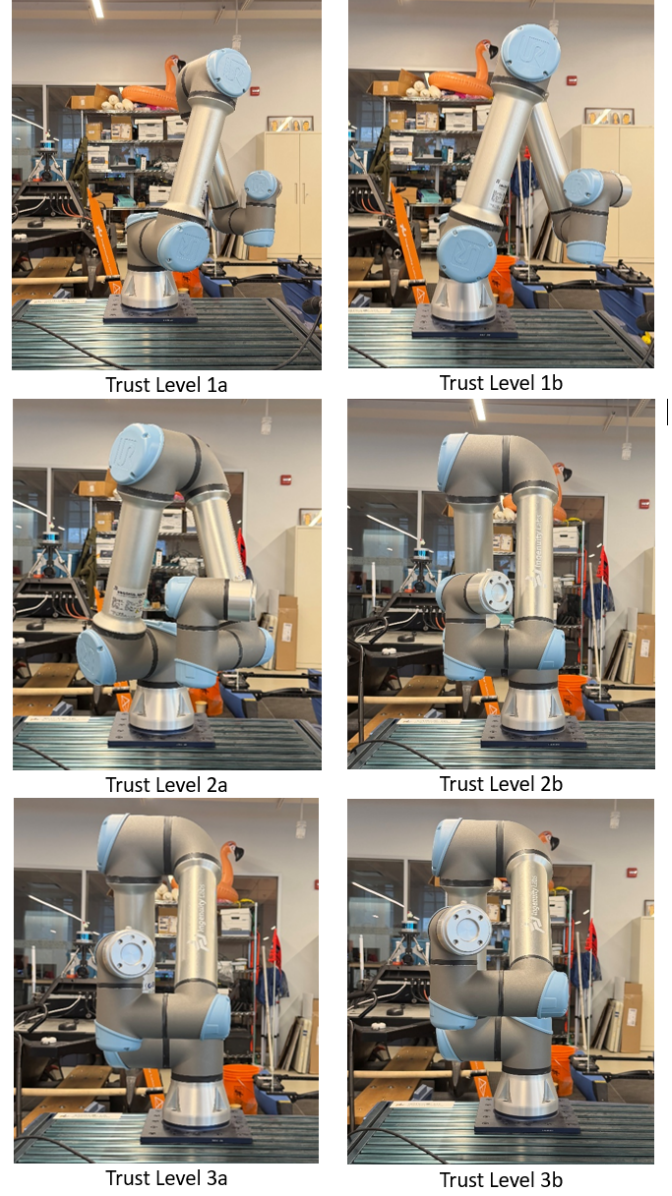


Fig. 2: Base configuration at different trust levels.

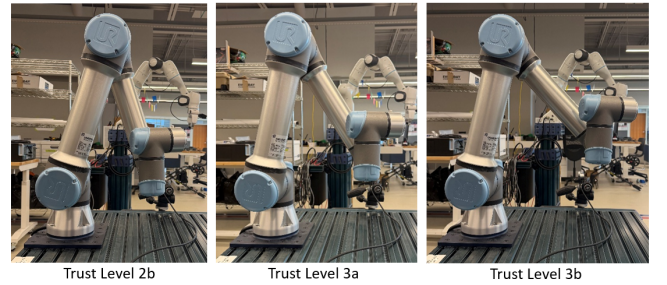


Fig. 3: Side view of UR5e change in posture during trust levels 3a and 3b, with comparison to trust level 2b.

Movement Parameters: There are three movement parameters: breathing, speed, and shake. The breathing parameter

controls how heavy the character appears to be breathing. The speed parameter controls how large a flinch the character computes when scared. The shake parameter determines the amplitude of the robot's shake or tremble. As trust levels increase, the movement parameters decrease to display calmer emotional states. These movements contribute to the development of the behaviors described in Section IV-C.

C. Behavior Selection

Using the above parameters, seven different behaviors can be defined:

1. Stare: From the base position, wrists one and two are adjusted so the robot appears to be looking at the user; the duration lasts anywhere from six to ten seconds.

2. Glance Surroundings: Within a 40-degree range in any direction, the robot will turn its head as if checking its surroundings; the duration can last between 0.5 and 1 second.

3. Shake: For trust level 1a, a sinusoidal offset is added to the Shoulder, Elbow, Wrist 1, and Wrist 2 joints. Trust levels 1b and above add the offset only to Wrist 1 and Wrist 2.

4. Tilt Head: This adds a 90-degree turn onto the Wrist 3 joint to simulate curiosity - similar to a dog tilting its head. This behavior only occurs when the robot is staring at the user.

5. Turn Base: This angles the robot Base joint toward the user during the interaction to simulate realistic adjustment.

6. Bounce: A small bouncing motion generated by applying a sinusoidal offset to the Elbow joint. This behavior is only performed while the robot is staring at the user.

7. Dance: A rhythmic motion generated by applying sinusoidal offsets to both the Elbow and Base joints. Like the bounce behavior, it is only performed while the robot is staring at the user.

Reactions: If any normalized limb velocity is above 0.5 or a hand is raised above the shoulder, the robot will switch to looking at the limb instead of the head. If user behavior results in a reduction of trust during levels 1a through 2a, the robot will exhibit a scared behavior: this includes a flinch away from the user and a shake that will last anywhere from three to six seconds. During this period, trust will not accumulate.

The probability of behavior depending on the trust level is shown in Table I.

TABLE I: Behavior selection probabilities (%) across trust levels.

Behavior	Trust Level					
	1a	1b	2a	2b	3a	3b
Stare	90	80	70	70	60	60
Glance at surroundings	10	20	30	30	40	40
Shake	100	100	0	0	0	0
Tilt head	0	0	30	60	40	0
Turn base	0	0	0	30	60	80
Bounce	0	0	0	0	30	50
Dance	0	0	0	0	0	30

The behavior selection probabilities were designed to reflect the progression of Interspecies Relationship Development, outlined in Section III. During the initial trust stage, behaviors

associated with vigilance and fear (observation and shaking) are strongly favored. As trust increases, behaviors indicative of curiosity (head turning and body orientation) become more probable. In the highest trust stage, expressive idle motions such as bouncing and dancing are introduced to represent relaxed and positive engagement [18].

V. EXPERIMENT DESIGN

This section outlines a potential user study to evaluate whether reciprocal trust development improves users' perceptions of and connection with the robot..

A. Conditions

Baseline: The robot displays the behavior associated with trust level 3b throughout the interaction. The UR5e arm maintains a low posture and controlled movement while remaining attentive and reactive to user motion. It occasionally performs positive idle behaviors such as bouncing and dancing.

Dynamic Trust: The robot transitions through all trust levels (1a–3b) over the course of the interaction.

B. Procedure

Since trust develops over time [18], participants would be randomly assigned to either the baseline or dynamic trust condition and would complete three interaction sessions over the course of three consecutive days.

Prior to the first interaction, users would complete the General Attitudes Towards Robots Scale (GAToRS) [21] to determine user attitudes toward robots. These measurements allow us to determine their predispositions about robots, which could affect experiment results.

To mimic an environment that may house a scared animal, the robot would remain in an enclosure. An informational sign would describe the robot's behavioral state and interaction guidelines. For the baseline condition, the sign would describe the robot as calm and content. For the dynamic trust condition, the sign would describe the robot as avoidant and scared of people. Both would direct readers to not touch the robot. Participants will be guided to read the paper, then enter the enclosure to interact.

Following each interaction, the user will complete the Godspeed Questionnaire Series (GQS) [22] and Robot Intimacy Receptivity Scale (RIRS) [23]. GQS measures anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety to understand user perception. RIRS measures communication, social distance, and personal intimacy, allowing us to evaluate the perceived relationship between the user and the robot. Both questionnaires are rated using five-point Likert scales, and scores for each dimension would be compared between the baseline and the dynamic trust condition. We hypothesize that dynamic trust will score higher in likeability, animacy, perceived intelligence, and interpersonal closeness.

VI. DISCUSSION

Creating interactions that are meaningful to the user is essential for successful HRI. Social experiences are intrinsically rewarding; people derive satisfaction from being accepted by others [24]. Trust is particularly meaningful because it requires one party to accept vulnerability in the expectation of a positive outcome. In addition, being trusted signals that another agent perceives the individual as safe, reliable, and valuable [25]. This perception of relational value contributes to a person's sense of meaning and self-worth, motivating people to seek relationships in which they feel valued by others [26].

The behavioral progression changes the user's role in the interaction. Rather than passively observing the robot, users must adjust their movement in response to the robot's reactions. Thus, the interaction becomes reciprocal, with both agents influencing the development of trust. Contrary to most robot applications, vulnerability is framed as a social behavior of the robot rather than a design limitation. Instead of immediately appearing confident or approachable, the robot initially communicates uncertainty and caution.

This interaction may also have applications beyond social robotics research. Because this interaction is reciprocal, it provides participants with immediate and consistent feedback about how their actions influence another agent's perceived comfort. Such interactions may support future educational systems that teach concepts such as respecting personal boundaries, interpreting non-verbal communication, and adapting behavior in response to social cues. Future work could investigate whether this approach is beneficial for children with ASD, where behavioral feedback may provide opportunities to practice social communication in a controlled environment.

A. Limitations and Future Work

The current trust model relies primarily on body pose, motion, and proximity to estimate user intent. These elements provide a strong indication of physical behavior; they do not capture other important social signals such as facial expression, eye contact, or noise. Incorporating facial recognition and audio reaction could enable more accurate estimation of user intention.

In addition, the trust model is based on manually selected thresholds and behavior probabilities that were tuned during development. Although this provides predictable behavior, future work could investigate adaptive approaches to personalize trust development or learn trust parameters directly from interaction data.

VII. CONCLUSION

This work presents a new approach to HRI centered on the dynamics of trust formation. Rather than designing robots to appear immediately trustworthy, we explore how trust can emerge through interaction. Our methodology incorporates animal-inspired behaviors and mannerisms through NARs to produce naturalistic social behaviors that encourage gradual relationship development. The proposed architecture serves as a foundation for future investigations into how these behaviors

influence perceived trust, social connection, and long-term human-robot bonds.

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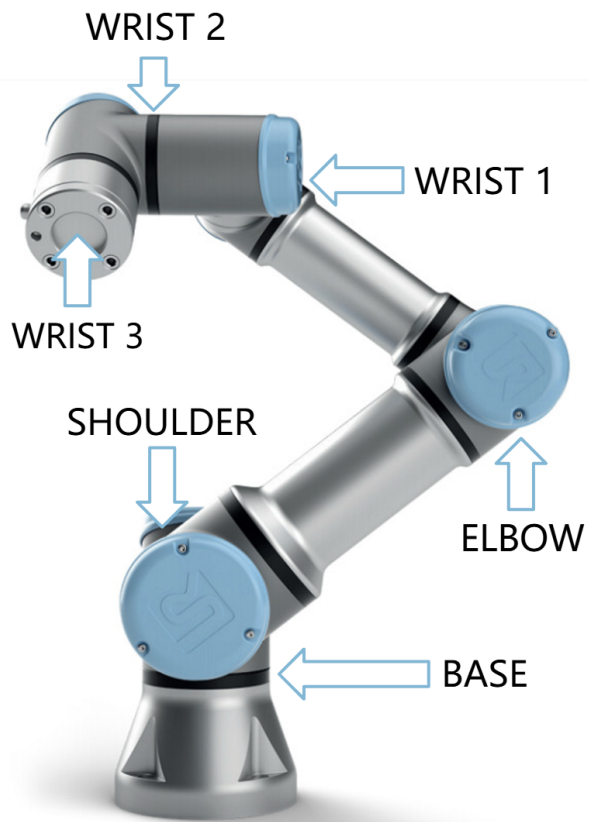


Fig. 4: Joint labels.